

Implementation Guide – PUBLIC
Version 1.3 (2025-06-21)

How to Integrate with SAP Business Network Freight Collaboration (Carrier APIs)

Implementation guide for B2B API integration

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INTRODUCTION

SAP Business Network for Logistics provides a network for shippers and carriers to collaborate for freight quotation, order confirmation, tracking, invoice, disputes and dock appointment use cases. SAP Business Network for Logistics provides web UI access to carriers for users to act on a defined process. Additionally, if you wish to get these documents in your systems, you may use the public APIs of SAP Business Network for Logistics to get the data and act on it. This document provides information on how to connect to SAP Business Network for Logistics as a carrier using REST APIs.

The following subset of processes are supported via APIs:

- Freight Tendering (Road)
- Freight Subcontracting (Road)
- Freight Booking (Ocean and Air)
- Order Tracking
- External Planning
- Settlement

For more general information, see the carrier introduction presentation and the carrier introduction video.

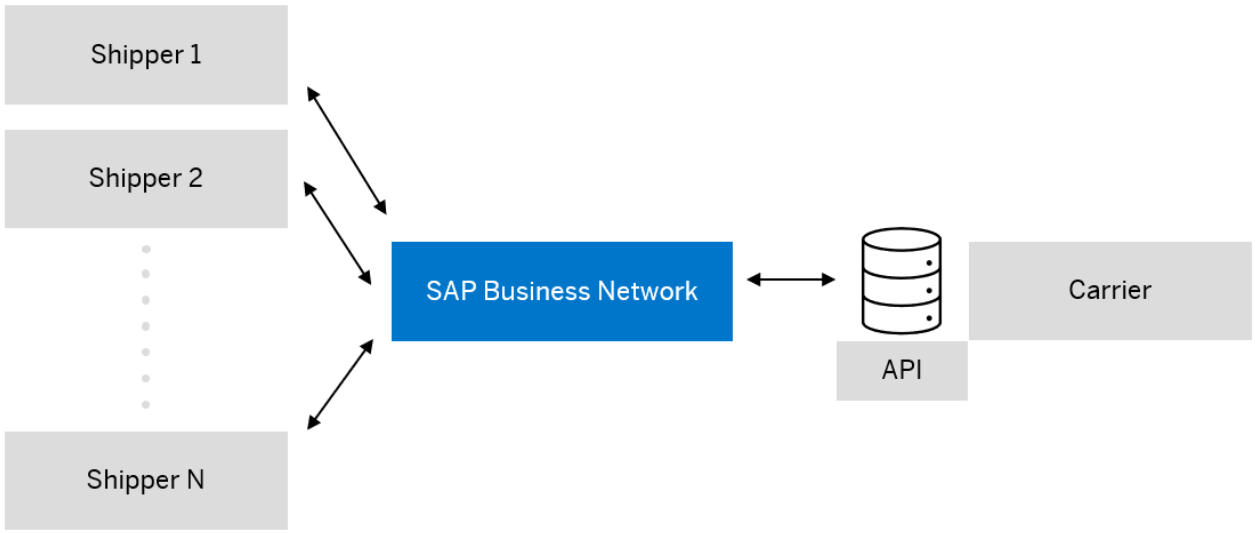
Note: All activities mentioned in this guide have to be done in both the test and productive environments.

PREREQUISITES

To set up API integration with SAP Business Network for Logistics, you must meet the following requirements:

- Carrier tenant is available in the test and productive environments. For more information about the process of registering SAP Business Network for Logistics, activating your account, logging on, and adding users, see the Activate Your Account guide.
- In case your API server uses IP filtering, you need to add the IP ranges to the allow list as mentioned in the administration guide for logistics providers, to enable SAP Business Network for Logistics integration tenants to connect to your API server. Please make sure the API server is accessible on the internet**.**
- It is expected that a carrier sets up an API server on their side for information exchange. Carriers have to set up the API connection once and shippers can use the same API connection for communication.

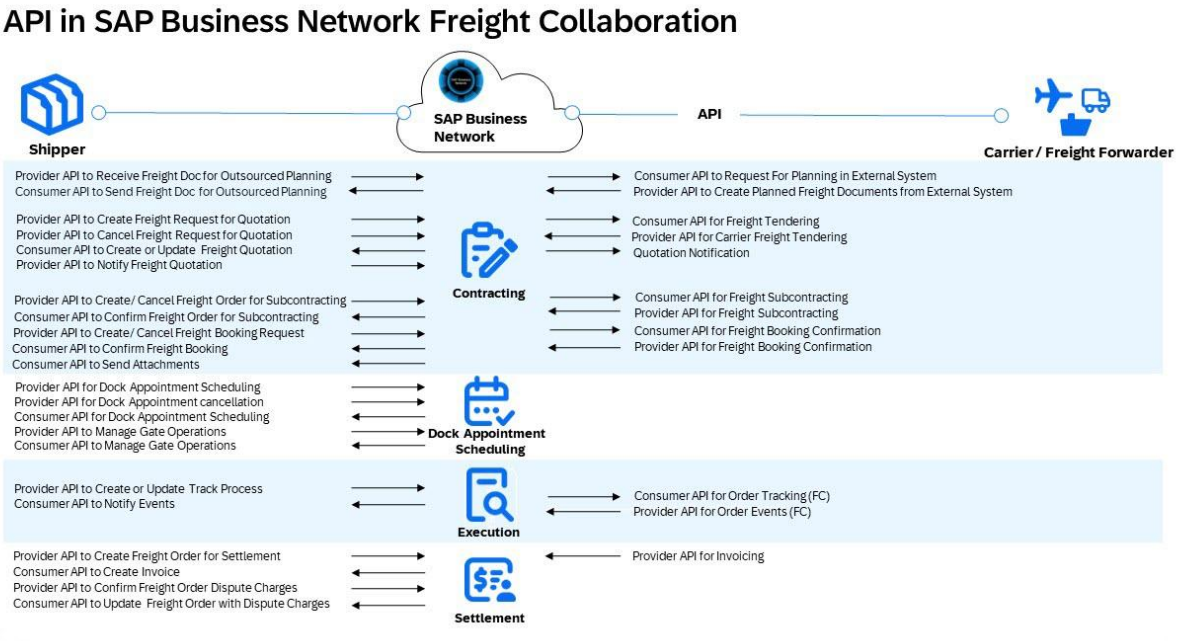
  The following diagram outlines the API communication between the involved parties:



API OVERVIEW

All APIs between the shipper and SAP Business Network Freight Collaboration are SOAP. All APIs between SAP Business Network Freight Collaboration and the carrier are REST.

SAP Business Network Freight Collaboration provides APIs for the following B2B processes:



You can find all the carrier-facing REST APIs on the right-hand side of the image.

For more information, see the following documents:

- Processes in SAP Business Network Freight Collaboration
- Carrier APIs (Rest APIs) in SAP Business Network Freight Collaboration

MIGRATION OF INTEGRATION PLATFORM

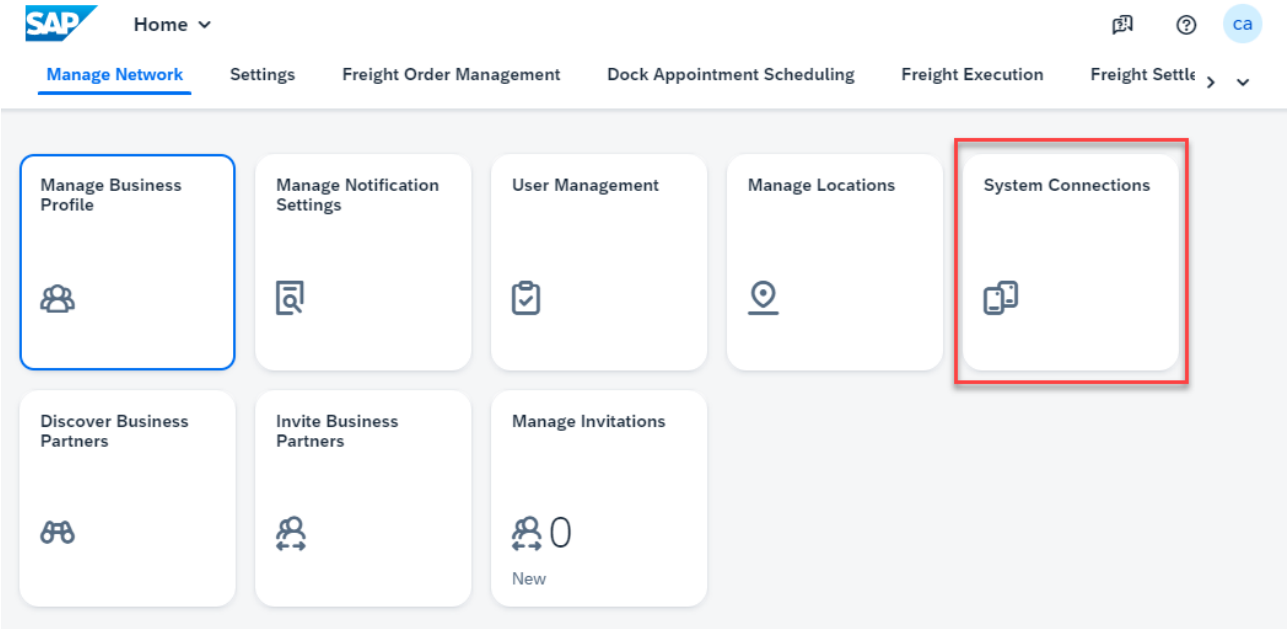
The integration platform used in SAP Business Network for Logistics is announced for sunset, and a new platform is enabled for customers and partners. If you are an existing customer who has setup API connectivity with the network, there are certain migration efforts required from your end. There's a detailed migration guide explaining the necessary steps required. We urge you to migrate to the new platform as soon as possible. For new customers, we've updated the endpoints and other details in our official product documentation on SAP Help Portal, the SAP Business Accelerator Hub documentation and also in this guide.

For more information, see the Integration Platform Migration Guide for Freemium Users of SAP Business Network for Logistics relevant for carriers.

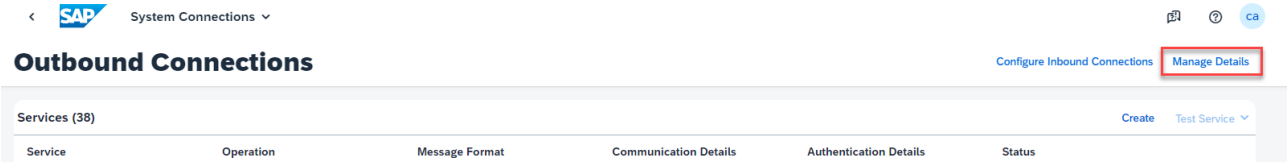
STEP 1: CONFIGURE OUTBOUND CONNECTIONS

Precondition: You have developed an API service in your backend server/transportation management system that can receive messages from SAP Business Network for Logistics in the published API format.

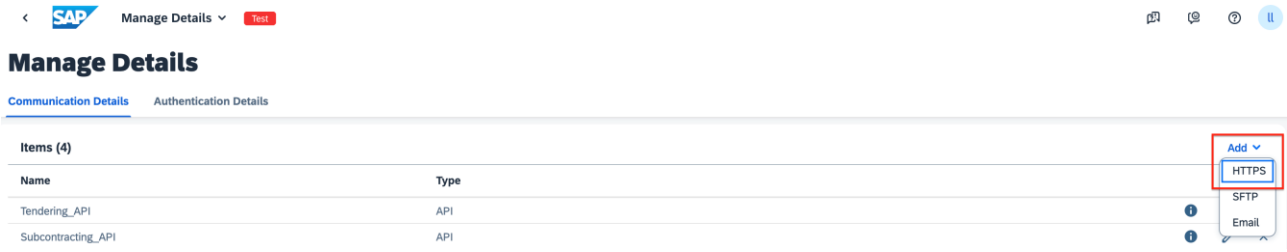
On the *Manage Network* tab of your SAP Fiori launchpad, select the *System Connections* app.



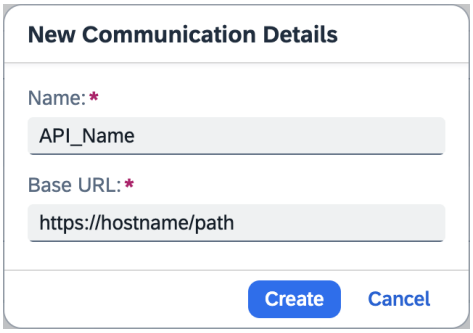
Choose *Manage Details* in the upper right corner.



Under the *Communication Details* tab, choose *Add* and select the HTTPS option from the dropdown menu.



Enter the following data in the *New Communication Details* window:



For more information, see Specifying the Authentication Details and Base URLs for Your System's Endpoints.

Special Considerations While Maintaining Communication Details

Due to disparity in the API designs of earlier SAP Business Network for Logistics APIs, there's a difference in how destination URLs need to be configured for different scenarios.

Earlier designs for 'Consumer API for Freight Subcontracting' and 'Consumer API for Freight Tendering' capture several pieces of information, such as shipper LBN ID, originator system ID and document ID, in the URL path especially for update, notification and cancellation scenarios. Hence the communication detail only expects the base URL to be maintained for all operation types (including create) of these two APIs. The URL path captured in the standard API documentation will be appended to the base URL configured in the communication details.

In the more recent API designs, such as tracking and freight booking, a different approach is followed. All necessary information is captured in the request body. Hence no URL path calculation is required. This means that the entire URL can be configured in the communication details. SAP Business Network for Logistics reads this configuration as-is, and no change is done while sending the request.

The difference in handling of communication details during runtime of different APIs is captured below:

Consumer API	**Communication Details - Base URL/Entire URL**
Freight Tendering	Base URL
Freight Subcontracting	Base URL
Freight Booking for Confirmation	Entire URL
Order Tracking	Entire URL
Request for Planning in External System	Entire URL

Examples:

Freight Subcontracting create scenario: If your API system is configured to accept a POST request at 'https://carrier-api-system/api/v1/LbnFreightOrderForSubcontracing', you need to capture the URL in the communication details as 'https://carrier-api-system/api/v1/'.

Freight Booking create scenario: If your API system is configured to accept a POST request at 'https://carrier-api-system/api/v1/FreightBookingRequest', you need to capture the entire URL in the communication details.

Authenticating Outbound Requests from Network

SAP Business Network for Logistics enables carriers to configure either a network-to-network authentication or a shipper-specific authentication for API request from the network to your backend system.

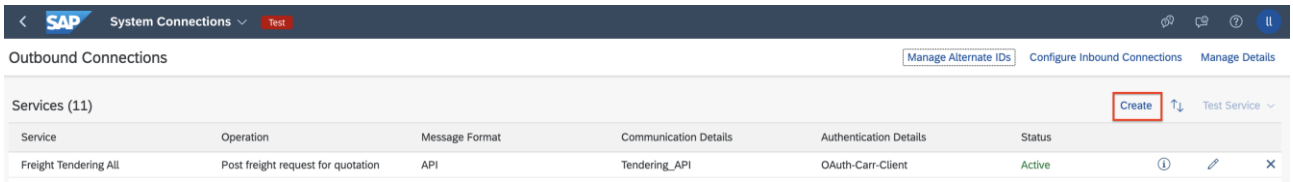
- Network-to-network – common authentication for all shippers in the network
- **Shipper-specific – you expect a unique authentication credential for requests sent by each shipper from the network**

If you wish to set up a network-to-network authentication, you can follow the steps below.

Under the *Authentication Details* tab, choose *Add* and select the required option from the dropdown menu and provide the credential details.

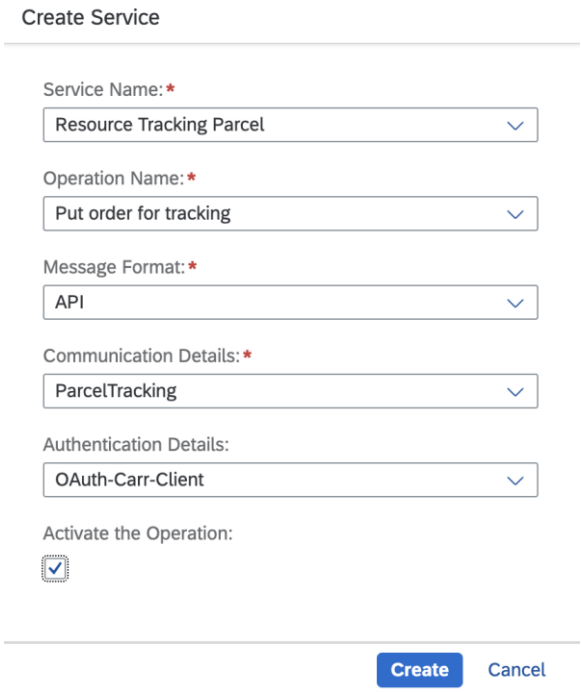


Go back to the entry page of the *System Connections* app and create outbound connections for all API services that you provide:

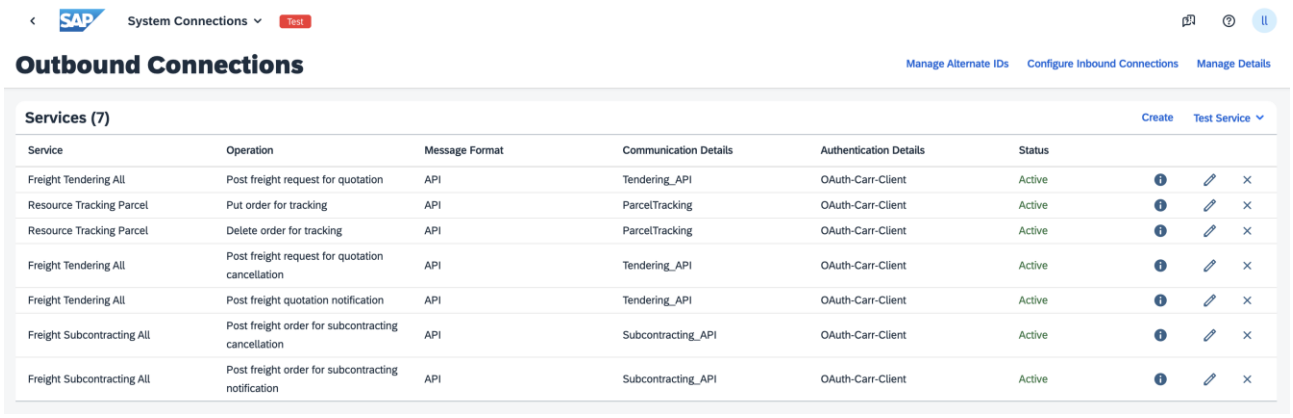


For a list of available services, see Assigning Authentication Details and a Base URL to a Service.

Enter the following data for the new service:

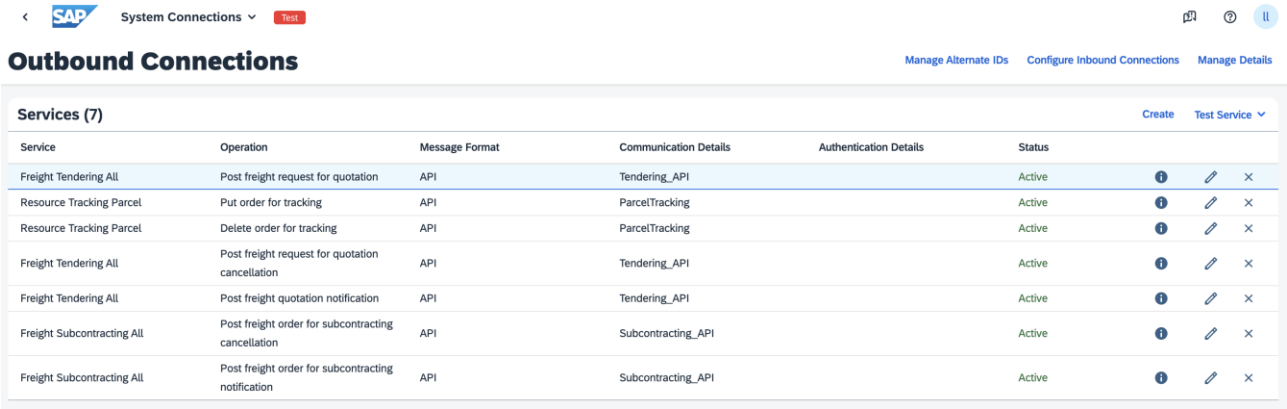


Please note that you need to maintain an “Authentication Detail” only if your integration design assumes a network-to-network authentication, this is, if you have a single authentication for all the shippers in SAP Business Network for Logistics.



[Image: *Configuration where carriers have a network-to-network authentication with SAP Business Network for Logistics*]

If you expect each shipper to send a unique authentication to your API system, you can leave this field blank, and shippers have a different configuration application where they'll provide the same.

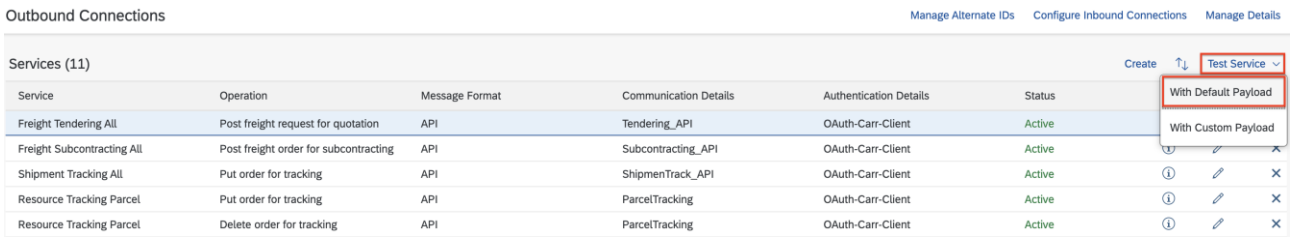


Service	Operation	Message Format	Communication Details	Authentication Details	Status
Freight Tendering All	Post freight request for quotation	API	Tendering_API		Active
Resource Tracking Parcel	Put order for tracking	API	ParcelTracking		Active
Resource Tracking Parcel	Delete order for tracking	API	ParcelTracking		Active
Freight Tendering All	Post freight request for quotation cancellation	API	Tendering_API		Active
Freight Tendering All	Post freight quotation notification	API	Tendering_API		Active
Freight Subcontracting All	Post freight order for subcontracting cancellation	API	Subcontracting_API		Active
Freight Subcontracting All	Post freight order for subcontracting notification	API	Subcontracting_API		Active

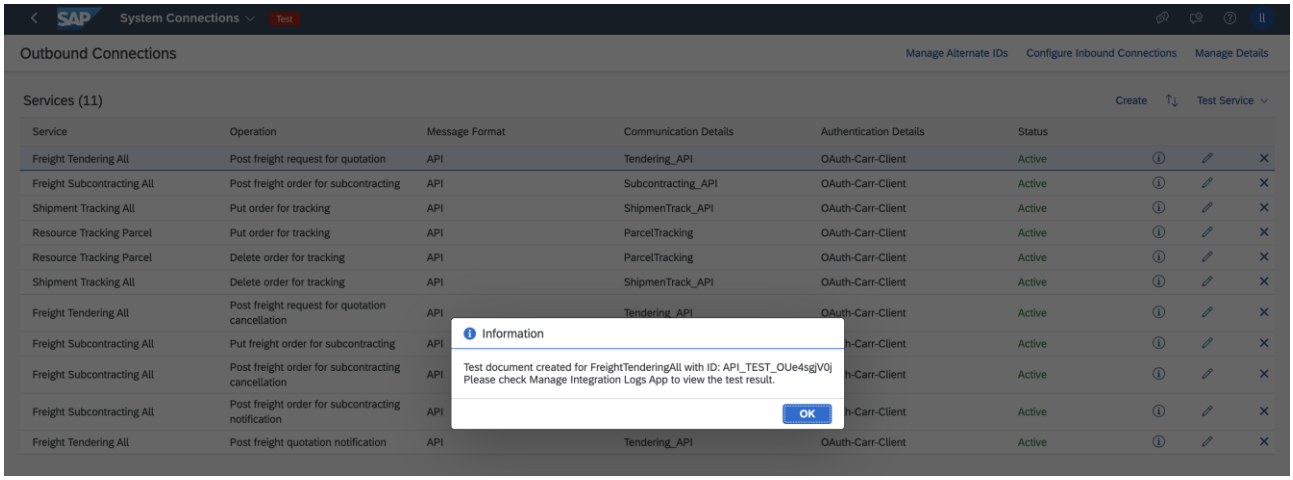
[Image: *Configuration where carriers expect a shipper-specific authentication from SAP Business Network for Logistics*]

STEP 2: TEST OUTBOUND CONNECTIVITY

After configuration, you should directly perform a connection test to your API server. To test the service, in the *Services* table, select *Post freight order for subcontracting* and choose *Test Service* in the upper-right corner, then select *With Default Payload*:



The system generates a test freight order and attempts to send a call to your API server. The below window is shown with the test freight order ID:



Using the ID of the created document (for example API_TEST_xVdlBk8sz6), open the *Manage Integration Logs* app to view the test result. Please refer to the chapter OPERATIONAL MONITORING for more details.